Aryan Mehta

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Education

Jaypee Institute of Information Technology — Noida, U.P
Bachelor of Technology in Computer Science | ***CGPA: 7.57*** — *Aug. 2024 – Present*

New Beersheba Senior Secondary School — Pithoragarh, U.K
PCM + Computer Science | ***93.3%*** — *Apr. 2021 – May 2023*

Experience

InAmigos Foundation — Remote
Data Analytics Intern — *May 2026 – July 2026*

- Compiled structured datasets on 15+ volunteer opportunities and 10 NGOs, using AI tools to synthesize social impact trends into strategic growth reports.
- Built a synthetic data pipeline (Python, pandas) as a self-directed follow-up project, simulating realistic NGO data-quality issues since real organizational data wasn't accessible.
- Applied PCA and K-Means clustering to segment 28 states into 3 empirically-justified performance tiers.
- Deployed an interactive Streamlit dashboard with live KPI rows and statistical hypothesis testing, backed by a 39-test automated pytest suite.

Achievements

- **Code Clash – Rapid Programming Hub**: 2x 1st Place Champion ('25 & '26) | 2nd Place, 26.1 | 5th Place, 25.2
- **3rd Place – OSDHack 2025**, OSDC
- **Google Cloud Arcade** – Arcade Trooper Tier, Core GCP Services (Compute Engine, Storage, IAM)

Projects

Redis Server Implementation | *C++17, POSIX Sockets, Multi-threading* — Jul. 2026

- Programmed a multi-threaded Redis server in C++17 utilizing raw POSIX sockets, deploying a thread-per-client concurrency model with fine-grained mutex locking.
- Constructed a master-replica replication system supporting the `PSYNC` handshake, asynchronous command propagation, and synchronous acknowledgment via `WAIT`.
- Designed a custom streaming parser for the Redis Serialization Protocol (RESP) and an RDB loader to deserialize persistent binary snapshots into memory.

OpenSpec – Self-Healing Accessibility CLI | *Node.js, axe-core, Llama 3, GitHub Actions* — Mar. 2026

- Created an accessibility CLI that executes WCAG 2.1 audits in a headless environment by loading HTML into a virtual DOM via jsdom and executing axe-core rules.
- Engineered a low-temperature prompting pipeline using Llama 3 via the Groq API to concurrently generate and apply semantically correct HTML fixes for failing nodes.
- Packaged the tool as a custom GitHub Action, enabling automated CI/CD gating to block deployments containing unresolved accessibility violations.

NGO Program Impact & Volunteer Analytics | *Python, pandas, scikit-learn, Streamlit* — Jun. 2026

- Architected a 6-stage data pipeline using pandas to clean and engineer features for 240+ synthetic volunteer records across 28 states, backed by 39 pytest checks.
- Implemented PCA and K-Means clustering to classify states into 3 performance tiers, deploying an interactive Streamlit dashboard for live KPI monitoring.
- Trained an exploratory logistic regression model using scikit-learn with balanced class weights to identify volunteer attrition risk factors based on engagement metrics.

Technical Skills

Languages: C++, Java, C, Python, JavaScript (Node.js), MySQL, HTML, CSS, Bash
Competitive Programming: Codeforces (Pupil), CodeChef (3⋆), HackerRank (5⋆), 500+ Problems (CodingNinjas)
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Multithreading, OS
Libraries: STL (Standard Template Library), NumPy, Pandas, Scikit-learn
Developer Tools: Linux (5+ years), Git, GitHub, Google Cloud Platform, VS Code, NeoVim